Data Architecture and Source Discovery Strategy for 'Fessi': A Comprehensive Framework for Circular Waste Management Knowledge Graphs

Executive Summary

The transition from a linear "take-make-dispose" model to a circular economy relies fundamentally on the flow of high-fidelity information. For a university environment, where the waste stream ranges from consumer electronics to specialized laboratory equipment, the challenge of proper disposal is not merely logistical but informational. The 'Fessi' project, envisioned as a knowledge graph powering a chatbot, represents a critical intervention in this space. By bridging the gap between users—who often lack the technical knowledge to identify hazardous components—and the complex regulatory frameworks governing waste, Fessi serves as a semantic translation layer. This report provides an exhaustive analysis of the data landscape required to build such a system, specifically tailored to the German context (ElektroG) and the EU WEEE Directive.

Our research identifies a fragmented data ecosystem. While "how-to" knowledge for disassembly is abundantly available through crowd-sourced platforms like iFixit, the authoritative "compliance" data regarding hazardous substances (SVHCs) is sequestered in regulatory databases like ECHA SCIP or static manufacturer PDFs. The primary challenge for Fessi is not data availability, but data integration. We propose a hybrid ingestion strategy that harmonizes structured API data with unstructured document scraping to populate a graph ontology grounded in the European Waste Catalogue (EWC).

This document details the technical specifications, legal constraints, and integration potential of key data sources, ranging from the iFixit API and the Open Repair Data Standard to manufacturer-specific portals (Apple, Dell, HP) and municipal waste APIs (MyMüll). It further outlines the ontological structures required to link a physical product (e.g., "iPhone 12") to its constituent hazards (e.g., "Lithium-ion battery") and its final legal disposal route (e.g., AVV 20 01 35*).

1. Introduction: The Information Asymmetry in E-Waste

The management of Waste Electrical and Electronic Equipment (WEEE) is governed by a complex web of material science and legal mandates. For a university student or facility

manager holding a broken device, the immediate questions are practical: "Can I fix this?" and "If not, which bin does it go in?" The answers, however, depend on invisible attributes: the chemical composition of the motherboard, the presence of a mercury backlight, or the type of battery embedded within the chassis.

1.1 The Regulatory Imperative

In Germany, the *Elektro- und Elektronikgerätegesetz* (ElektroG) transposes the EU WEEE Directive (2012/19/EU) into national law. A critical component of this regulation is **Article 15**, which mandates that producers provide information free of charge about preparation for re-use and treatment for each type of EEE placed on the market.[1] This "Information for Recyclers" is intended to identify components and materials that require selective treatment—specifically those listed in **Annex VII** of the Directive.

For Fessi, Article 15 is the "North Star." Every data source evaluated in this report is assessed against its ability to answer the Annex VII question: *Where are the hazards located, and how do we remove them?*

1.2 The Knowledge Graph Approach

A traditional relational database struggles with the heterogeneity of waste data. A knowledge graph, however, excels at mapping these relationships. The target ontology for Fessi must capture:

- **The Entity:** The physical device (Brand, Model).
- **The Component Structure:** The mereology of the device (HasPart -> Battery).
- **The Procedure:** The sequence of actions required to disassemble the entity (Step 1, Step 2...).
- **The Classification:** The mapping of the entity or component to a legal waste code (AVV).

The following sections analyze the specific data repositories available to populate these four pillars.

2. Community and Crowd-Sourced Repair Repositories

The most granular source of disassembly logic—the actual "how-to" instructions required to separate components—lies not with manufacturers, but with the global repair community. These platforms utilize crowd-sourcing to build vast libraries of repair guides that are structurally ideal for graph ingestion.

2.1 The iFixit API Ecosystem

iFixit is the de facto standard for open repair documentation. Its data structure is inherently

semantic, organizing information into a hierarchy of Category -> Device -> Guide -> Step. This mirrors the desired structure of the Fessi graph.

2.1.1 API Architecture and Endpoints

The iFixit API (currently v2.0) provides RESTful access to the platform's content.[3] For a waste management knowledge graph, the following endpoints are critical:

A. The Category Hierarchy (GET /categories)
This endpoint returns the taxonomic tree of devices. In the context of a knowledge graph, these categories serve as the class hierarchy (e.g., Electronics -> Computer -> Laptop -> MacBook).

- **Relevance:** It allows Fessi to infer properties. If a device is classified under "Laptops," the graph can infer a high probability of containing a Lithium-ion battery and a Screen, triggering specific waste disposal queries.
- **Filtering:** The API supports parameters like requireGuides=true [3], ensuring that the graph is only populated with devices for which actionable disassembly data exists. This filters out "stub" pages that would clutter the database without adding utility.

B. The Guide Interface (GET /guides/{id})
This is the core data source for disassembly logic. A "Guide" in iFixit's schema is a sequence of steps.

- **Step Data:** Each step contains text instructions (text_raw), images, and associated metadata.[4] This granularity is vital. A step text reading "Pry up the battery connector" allows the graph to link the Battery node to a DisassemblyAction.
- **Tool Dependencies:** The guide payload lists required tools (e.g., "Pentalobe Screwdriver"). In a university makerspace or recycling center context, this data informs the user not just *what* to do, but *whether they can* do it.
- **Prerequisites:** Guides are linked via prerequisite relationships.[5] For example, a "Battery Replacement" guide might list "Case Removal" as a prerequisite. This dependency graph is essential for calculating the total effort and complexity of a teardown operation.

C. Device Metadata (GET /devices/{id})
Beyond instructions, this endpoint provides specifications. While often sparse, it can contain data on release dates and dimensions, helping to distinguish between generations of hardware (e.g., iPhone 12 vs. iPhone 13) which may have different hazardous material profiles.

2.1.2 Licensing and Utilization Constraints

Data sovereignty is a key consideration for university projects. iFixit content is licensed under **CC-BY-NC-SA 3.0** (Creative Commons Attribution-NonCommercial-ShareAlike).[6]

- **Implications:** Fessi can freely use this data for a non-commercial university chatbot. However, the "ShareAlike" clause implies that if Fessi enriches this data (e.g., by correcting a step), that improvement should ideally be contributed back or made available under similar terms.

- **Commercial Limitations:** If the chatbot were to be commercialized or sold to waste management firms, this license would pose a barrier. For the current scope, it is permissible and highly advantageous.

2.1.3 Integration Strategy: python-ifixit

To streamline ingestion, the python-ifixit library (and related tools like ifixit2zim) provides a wrapper around the API.[7]

- **Functionality:** It handles pagination and can dump entire categories.
- **Search Optimization:** The API includes a /suggest endpoint.[6] This is crucial for the chatbot's NLP layer. When a student types "How do I trash my MacBook?", the system first hits /suggest to resolve "MacBook" to a specific Device ID, then fetches the relevant guides.

2.2 The Open Repair Data Standard (ORDS)

While iFixit tells us *how* to repair, the Open Repair Alliance tells us *what* is breaking. The Open Repair Data Standard (ORDS) is a schema designed to capture the outcomes of community repair events.[8]

2.2.1 Schema Analysis

The ORDS v0.3 specification defines fields that are highly predictive for waste management [10]:

- **product_category**: A controlled vocabulary for devices, which can be mapped to the iFixit hierarchy and EWC codes.
- **fault_type**: Describes what went wrong (e.g., "Battery failure", "Screen broken").
- **repair_status**: Indicates if the device was "End of Life" or "Repairable."
- **repair_barrier**: Records why a repair failed (e.g., "Spare parts too expensive").

2.2.2 Predictive Intelligence for Fessi

By ingesting the ORDS datasets (hosted on GitHub), Fessi can move beyond static instructions to predictive advice.

- **Scenario:** A user asks about a generic "broken toaster."
- **Data Inference:** Fessi checks ORDS statistics for toasters. If 60% of toaster failures are due to "heating element failure" (non-hazardous) and 10% are "PCB failure" (potentially hazardous), the chatbot can tailor its triage questions: "Is the heating element broken, or does it not turn on at all?"
- **Source:** The data is available as CSV/JSON downloads from the openrepair/data repository.[12] This "batch" data complements the real-time API data from iFixit.

3. Manufacturer Compliance Data: The Authoritative

Source

While community data is rich in detail, manufacturer data is rich in authority. For WEEE compliance, the manufacturer's official designation of hazardous components is the legal standard. However, this data is often locked in non-machine-readable formats.

3.1 Apple Recycler Guides

Apple represents a "best case" for manufacturer transparency in the context of WEEE. The company publishes dedicated "Recycler Guides" for most products, specifically to comply with the WEEE Directive.[13]

3.1.1 Document Taxonomy

These guides are distinct from consumer manuals. They are technical documents aimed at treatment facilities.

- **Scope:** They cover specific models (e.g., iPhone 17, MacBook Air, iPad Pro).[14]
- **Key Section: Annex VII Components:** Every guide contains a table explicitly mapping **Directive 2012/19/EU Annex VII** requirements to the device's components.
    - *Example:* For a MacBook Air, the table identifies the "Main logic board" (Printed circuit board >10cm²) and "Lithium-ion polymer batteries" as requiring separate treatment.[15]
- **Disassembly Flow:** The guides provide linear, step-by-step instructions (e.g., "Remove the back glass by hand," "Pry off the main logic board").[14]

3.1.2 The "PDF Gap" and Scraping Strategy

Unlike iFixit, Apple does not offer an API for this data. It is hosted as static PDFs. To populate Fessi, a specialized ingestion pipeline is required.

- **Tools:** Repositories like apple-documentation-archive-scraper or generic PDF extraction libraries (smalot/pdfparser) must be employed.[17]
- **Target Extraction:** The parser should focus on the "Annex VII" table. This table is structurally consistent across documents. Extracting rows where "Removal Instructions" are linked to component names allows Fessi to flag high-priority disassembly steps.
- **Hybrid Verification:** Fessi can use Apple's data to *validate* iFixit guides. If Apple lists a "Taptic Engine" as containing rare earths [19], Fessi can ensure the corresponding iFixit guide highlights this component as a recovery target.

3.2 Dell and HP: Varied Approaches to WEEE

Other manufacturers adopt different strategies, necessitating flexible ingestion logic.

- **Dell:** Dell explicitly states that consumer user manuals do *not* contain disassembly instructions for safety reasons.[20] Instead, they provide "Recycler Information" sheets.

    - *Content:* These often focus on negative assertions (e.g., "Product does not contain Mercury") or high-level component identification.[21] They are less instructional and more declarative.
    - *Automation:* Dell provides APIs for "End of Service Life" (EOSL) and warranty data [22], and tools to export hardware inventory to JSON.[23] While not disassembly instructions, this data helps Fessi identify the age and likely waste status of a Dell device.
- **HP:** HP's disassembly instructions are often more visual, providing exploded views for recyclers.[24] These documents are crucial for identifying capacitors containing PCBs or electrolytes, a specific Annex VII hazard.[25]

3.3 The I4R Platform (Information for Recyclers)

The fragmentation of manufacturer data led to the creation of the **Information for Recyclers (I4R)** platform, a centralized hub managed by the WEEE Forum.[1]

3.3.1 Role in the Ecosystem

I4R allows recyclers to access recycling information at the **product category level**.

- **Coverage:** It currently hosts ~47 product fiches across 6 categories.[2] While this seems low compared to iFixit's thousands of guides, the *category-level* data is a vital fallback.
- **Utility:** If Fessi encounters a rare washing machine brand with no specific manual, it can query the I4R data for "General Washing Machine" to retrieve standard hazard locations (e.g., capacitor banks, door locks).
- **Access:** The platform is designed for registered recycling companies. While there is no public API advertised in the snippets [2], the underlying data structure is harmonized. Fessi's developers should seek academic access or utilize the harmonized reporting formats (fiches) as a schema template for the graph.

4. Hazardous Material Intelligence: The "Deep" Data

Disassembly is only useful if we know *why* we are disassembling. The chemical composition of components determines their toxicity and value.

4.1 ECHA SCIP Database

The **Substances of Concern In articles as such or in complex objects (Products)** (SCIP) database is the EU's central repository for tracking Substances of Very High Concern (SVHCs).[27]

4.1.1 The SCIP Data Model

SCIP uses the **IUCLID** (International Uniform Chemical Information Database) format. This model is highly granular:

- **Complex Object:** The finished product (e.g., Laptop).
- **Article:** The component (e.g., O-ring, Chip).
- **Substance:** The chemical (e.g., Lead, Cadmium).[28]

4.1.2 Integration via System-to-System (S2S) API

ECHA provides a **System-to-System (S2S)** API for automated data exchange.[29]

- **Workflow:** Fessi can query this database using product identifiers (GTIN, EAN).
- **Payload:** The response includes the "Safe Use Instructions".[29] This text is gold for a chatbot. If a student asks about a specific lab instrument, Fessi can retrieve the SCIP dossier and warn: "Contains Lead in the chassis; wear gloves and do not incinerate."
- **Material Recovery:** SCIP data helps identify value. Knowing a device contains specific recoverable metals allows the chatbot to route it to "High Value Recovery" rather than "General E-Waste."

4.2 ProSUM and the Urban Mine Platform

The ProSUM project (Prospecting Secondary raw materials in the Urban mine and Mining wastes) offers macro-level data.[30]

- **Data:** It provides the average chemical composition of product groups (e.g., average Gold content in a smartphone).
- **Application:** This data powers the "Impact" logic of the chatbot. Fessi can tell a user, "Recycling this phone recovers 0.03g of Gold," adding a behavioral nudge to the compliance instruction.

5. Waste Classification: The German Legal Context

A German university chatbot must ultimately speak the language of German bureaucracy. The physical object must be mapped to a legal code.

5.1 The European Waste Catalogue (EWC) and AVV

In Germany, the **Abfallverzeichnisverordnung (AVV)** implements the European Waste Catalogue.

- **Structure:** A 6-digit code system (e.g., 16 02 14).
- **Hazardous Flag:** Codes marked with an asterisk (*) denote hazardous waste.[31]
- **Ontology:** The **ewc-onto** project translates this catalogue into an OWL ontology.[32]
    - *Implementation:* Fessi should import this ontology. When a product is identified as containing "Annex VII components" (via Apple PDF or iFixit), the graph logic should automatically classify the waste as hazardous (e.g., 20 01 35*) until those components are removed.

5.2 Municipal APIs: MyMüll and AbfallPlus

Waste management is local. The rules for "Yellow Bin" vs. "Recycling Center" vary by city.

- **MyMüll:** Used by many German municipalities. It offers an API (api.php) that accepts city_id and area_id to return collection schedules.[33]
- **AbfallPlus:** Another common provider. Its API uses a key (often retrievable from the municipal website's network traffic) to fetch street-specific data.[34]
- **Integration:** Fessi must ingest these to answer "When" and "Where." By connecting the AVV code (The "What") to the Municipal API (The "Where"), Fessi closes the loop for the user.

6. Knowledge Graph Architecture and Ingestion Strategy

To synthesize these disparate sources, Fessi requires a robust architecture. The following section outlines the proposed schema and ingestion pipelines.

6.1 The Fessi Schema (Ontology)

The graph should be built around a core set of entities that link the physical, procedural, and legal domains.

Entity Class	Description	Primary Data Source	Key Properties
Product	The physical device.	iFixit API, ECHA SCIP	Brand, Model, GTIN, Category
Component	A part of the product.	Apple PDFs, iFixit	Name, IsHazardous, MaterialComposition
DisassemblyStep	An action to remove a component.	iFixit API, Apple PDFs	InstructionText, ToolRequired, OrderIndex
WasteCode	The legal classification.	AVV (ewc-onto)	Code (e.g., 16 02 14), Description, IsHazardous

DisposalPoint	The physical location for waste.	Municipal APIs	LocationName, OpeningHours, AcceptedCodes

6.2 Data Ingestion Pipelines

Pipeline A: The Structured Harvester (iFixit & APIs)

1. **Poll GET /categories**: Build the taxonomy tree.
2. **Poll GET /guides**: For each relevant category, fetch guides.
3. **Entity Mapping**: Map iFixit "Tools" to a standardized Tool node. Map iFixit "Parts" to Component nodes.
4. **Hazard Flagging**: Use keyword heuristics (e.g., "Battery", "Mercury") in part names to set IsHazardous = True.

Pipeline B: The Document Scraper (Manufacturer PDFs)

1. **Acquisition**: Scripts fetch PDFs from known URLs (e.g., apple.com/recycling/recycler-guides/).
2. **Extraction**: Use pdfparser to locate the "Annex VII" table.
3. **Triangulation**: Match the model number in the PDF (e.g., "A2179") to the iFixit device.
4. **Enrichment**: Inject the high-fidelity hazard data from the PDF into the iFixit-derived graph structure, overwriting generic heuristics with authoritative data.

Pipeline C: The Compliance Layer (SCIP & AVV)

1. **SCIP Query**: For high-volume university assets (monitors, PCs), query the SCIP database via S2S API to get chemical data.
2. **AVV Mapping**: Use SWRL (Semantic Web Rule Language) rules to infer waste codes.
    - *Rule:* Product(?p) ^ hasComponent(?c) ^ IsHazardous(?c, true) -> hasWasteCode(?p, "20 01 35*")

7. Strategic Recommendations and Gap Analysis

7.1 Addressing the "Long Tail" of University Equipment

While iFixit covers consumer electronics well, university labs contain specialized equipment (centrifuges, spectrometers) that lacks public repair guides.

- **Recommendation:** Implement a "fallback" logic using I4R category data. If a specific "Beckman Coulter Centrifuge" guide is missing, default to the "Small Industrial Appliance" hazard profile (check for capacitors, heavy metals).
- **Crowdsourcing:** Allow university technicians to submit PDF manuals for lab equipment directly to Fessi, parsing them via the Pipeline B scraper.

7.2 Legal and Licensing Considerations

The reliance on iFixit (CC-BY-NC-SA) and scraped manufacturer data (Copyright) requires careful handling.

- **Attribution:** The chatbot must explicitly attribute iFixit for any disassembly steps displayed.
- **Right to Repair:** Leveraging these sources aligns with the EU's Right to Repair push. However, Fessi should prioritize "Official Recycler" information (Apple/Dell PDFs) for safety-critical hazards (e.g., swollen batteries) to mitigate liability.

7.3 Future-Proofing: The Digital Product Passport

The EU is moving toward a **Digital Product Passport (DPP)**. This will eventually standardize much of the data Fessi currently has to scrape.

- **Strategy:** Design the Fessi ontology to be compatible with the emerging DPP standards (based on GS1/EPCIS). The current schema of Product -> Component -> Material is already aligned with this future state.

8. Conclusion

Building 'Fessi' is an exercise in synthesizing the fragmented truth of the electronic product lifecycle. No single source provides the complete picture. **iFixit** provides the procedural "muscle" (how to take it apart), **Apple/Dell/SCIP** provide the regulatory "conscience" (what is dangerous), and **AVV/Municipal APIs** provide the local "reality" (where it goes). By integrating these streams into a coherent knowledge graph, the university can create a tool that not only ensures compliance with German law but actively educates its community on the mechanics of the circular economy. The proposed architecture—a hybrid of API harvesters and document scrapers feeding a semantic ontology—offers a robust, scalable path to achieving this goal.

Detailed Analysis of Research Snippets and Data Sources

The following sections provide a granular breakdown of the specific data points, API behaviors, and document structures discovered during the research phase.

2.1 iFixit API V2.0: Deep Dive

The iFixit API is the backbone of the "Disassembly Instruction" domain. Its granular data

model allows for precise extraction of steps, tools, and safety warnings.

Technical Specifications:

- **Base URL:** https://www.ifixit.com/api/2.0
- **Authentication:** Not required for read-only public data, but rate limits apply.
- **Response Format:** JSON.

Key Endpoints and Data Mapping:

Endpoint	JSON Structure Highlights	Graph Mapping Logic
/categories	{"name": "MacBook Pro", "unique_id": "...", "ancestors": [...]}	Maps to ProductCategory nodes. Ancestors build the subClassOf hierarchy.
/devices/{id}	{"title": "iPhone 12", "repairability_score": 6}	Maps to Product nodes. Repairability score helps triage "Repair vs. Recycle" decisions.
/guides/{id}	{"steps":}]}	Maps to DisassemblyProcess and Step nodes. Tools link to Tool entities.
/suggest	{"results":}	Used for NLP entity resolution in the chatbot.

License Nuance:
The content is CC-BY-NC-SA 3.0.6 This allows the university to use the data for an internal (non-commercial) chatbot but requires:

1. **Attribution:** Fessi must display "Source: iFixit" with a link to the original guide.
2. **ShareAlike:** If the university improves the data (e.g., translating steps to German), these improvements should ideally be shared.

Limitations:

- **Coverage:** Excellent for consumer IT (Laptops, Phones), poor for "White Goods" (Fridges, Washing Machines) and "Lab Equipment."
- **Localization:** While iFixit supports multiple languages, the availability of German guides

is lower than English. Fessi may need a translation layer (e.g., DeepL API) for the German-facing chatbot.

2.2 Open Repair Data Standard (ORDS)

The ORDS offers a standardized way to describe *repair attempts*. This data is valuable for the "Diagnosis" phase of the chatbot.

Schema Details (v0.3):

- **product_category:** Harmonized list (e.g., "Laptop", "Food Processor").[10]
- **fault_type:** Controlled vocabulary for defects.
- **repair_barrier:** Critical for identifying *why* things are discarded. If 80% of "Inkjet Printers" are discarded due to "Cost of parts," Fessi can suggest recycling immediately rather than attempting repair.

Data Source:

- **Repository:** https://github.com/openrepair/data.[12]
- **Format:** Aggregated CSV files.
- **Integration:** Periodic batch ingestion to update statistical properties of ProductCategory nodes in the graph (e.g., hasRepairSuccessRate).

3.1 Manufacturer Documentation: Apple

Apple's documentation is a gold standard for WEEE compliance because it explicitly addresses the **Annex VII** requirements.

Document Structure (PDF):

- **Title:** "Apple Recycler Guide".[13]
- **Target:** Professional Recyclers.
- **Annex VII Table:** A dedicated table listing:
  - *Substance/Component:* e.g., "Lithium-ion polymer batteries."
  - *Apple Part Name:* e.g., "Battery."
  - *Removal Instructions:* e.g., "Follow steps 1–10".[16]
- **Material Categorization:** Lists output fractions (e.g., "Mixed Electronics," "Battery").[19]

Specific Examples Found in Research:

- **iPhone 17:** "Remove the back glass by hand... remove the lithium-ion battery from the enclosure".[14]
- **MacBook Air (Retina):** Explicitly lists "Printed circuit board if surface > 10 sq cm" and "External electric cables" as requiring separate treatment.[15]
- **iPad Pro:** Identifies "Display logic board" and "Main logic board" as targets.[35]

Ingestion Strategy:
A targeted PDF scraper is necessary. The consistency of the "Annex VII" table layout across different guides makes this automatable. The extracted data provides the authoritative link between a product and its hazardous components, which is legally required for proper disposal.

3.2 Manufacturer Documentation: Dell & HP

Dell:

- **Approach:** Provides "WEEE Recycler Information" sheets.
- **Content:** Focuses on material presence. "Product does not use internal batteries based on Mercury... Product uses a Lithium Primary Coin Cell Battery".[21]
- **Automation:** APIs for inventory management (HardwareInventory) can export component lists to JSON [23], which can assist in identifying the configuration of university assets.

HP:

- **Approach:** Disassembly instructions often include specific references to identifying capacitors and batteries.[24]
- **Format:** PDFs similar to Apple, but often with more technical/exploded view diagrams.

4.1 ECHA SCIP Database

The **SCIP Database** is the central EU repository for chemical compliance.

API Details:

- **Type:** System-to-System (S2S) submission and retrieval.[29]
- **Format:** IUCLID (XML/JSON).[28]
- **Key Data:**
  - **SVHCs:** Identifies if a product contains Candidate List substances >0.1%.
  - **Article Hierarchy:** Links complex objects to simple articles.
  - **Safe Use:** Text fields providing handling instructions.[29]

Relevance:
For a university, this database is crucial for "Lab Safety." If a piece of equipment contains a carcinogenic substance (SVHC), Fessi must prioritize this information over general recycling advice. The S2S API allows Fessi to fetch this data dynamically for assets with known identifiers (GTIN/EAN).

5.1 European Waste Catalogue Ontology (ewc-onto)

To make the data machine-understandable, it must be mapped to an ontology.

The Ontology:

- **Source:** https://github.com/kschmidt-at-dbfz/ewc-onto.[32]
- **Format:** OWL (Web Ontology Language).
- **Structure:** It digitizes the EWC hierarchy (20 Chapters, 842 Waste Types).
- **Properties:** It captures the "hazardous" property (asterisk in AVV) as a formal attribute.

Usage:
Fessi uses this as its "Output Layer."
- Input: "iPhone 12" (Product)
- Process: Identify components -> Separate Battery -> Separate Mainboard.
- Output:
  - Battery -> 16 06 05 (Lithium batteries)
  - Rest of Phone -> 20 01 36 (Discarded equipment)
  - Screen -> 20 01 35* (If hazardous components present)

5.2 Municipal APIs (MyMüll / AbfallPlus)

MyMüll (JumoMind):

- **Endpoint:** https://mymuell.jumomind.com/mmapp/api.php.[33]
- **Parameters:** r=dates, city_id, area_id.
- **Output:** JSON list of collection dates and trash types.

AbfallPlus:

- **Endpoint:** https://api.abfallplus.de/index.php.[34]
- **Auth:** Requires a key (often found in web traffic of municipal sites).
- **Output:** Street-level schedules.

Integration:
These APIs allow Fessi to be context-aware. It transforms the advice from "You should recycle this" to "Take this to the recycling center on Main Street; it opens at 09:00."

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6. Conclusion and Future Outlook

The 'Fessi' project sits at the intersection of open data activism (iFixit), corporate compliance (Apple/Dell), and regulatory enforcement (ECHA/AVV). The research demonstrates that while a "universal API" for waste does not exist, the components to build one are available.

By leveraging the **iFixit API** for procedural knowledge, scraping **Manufacturer Guides** for compliance validation, and querying **ECHA SCIP** for chemical safety, Fessi can construct a comprehensive graph of the device lifecycle. Mapping this data to the **EWC Ontology** and local **Municipal APIs** ensures that this knowledge is not just theoretically accurate but practically actionable for the university community. The result is a system that turns the

opaque problem of "e-waste" into a transparent, navigable process, directly supporting the goals of the Circular Economy.

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